Table 7.11: Various toxins of *Shigelladysenteriae*

Toxins	Mode of Action
Endotoxin	It is released after autolysis, it has irritating effect on intestinal wall which causes diarrhea and subsequently intestinal ulcers.
Exotoxin	It is a powerful toxin and acts as Enterotoxin as well as neurotoxin. **As Enterotoxin** – It induces fluid accumulation **As Neurotoxin** – It damages the endothelial cells of small blood vessels of CNS which results in polyneuritis and coma
Vero cytotoxin	It acts on Vero cells

Site of infection – Large intestine

Incubation Period – Less than 48 hours (1–7 days)

Mode of transmission – Food, finger, faeces and flies

7.8.5 Clinical Manifestations

- Frequent passage of loose, scanty faeces containing blood and mucus.
- Abdominal cramps and tenesmus (straining to defecate).
- Fever and vomiting.
- Hemolytic uremic syndrome (It is a condition caused by the abnormal destruction of red blood cells).

Sh. dysenteriae causes bacillary dysentery. The pathogen enters into the host by the ingestion of contaminated food.

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The bacilli reaches large intestines and adheres to the epithelial cells of villi. It multiples inside the cell and penetrates into lamina propria.

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As the pathogen multiples it produces toxins, it stimulates an inflammatory reaction and causes extensive tissue destruction. It leads to necrosis of surface epithelial cells.

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The necrotic epithelia, become soft and friable and are sloughed off leaving behind transverse superficial ulcers.

↓

Abdominal cramps and pain are caused by the disruption of the muscular function of the intestine

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The degeneration of intestinal villi and local erosion causes bleeding, heavy mucous secretion resulting in bacillary dysentery

Flowchart 7.5: Pathogenesis of *Shigella dysenteriae*

7.8.6 Laboratory Diagnosis

Specimens: Fresh stool is collected.

Direct Microscopy: Saline and Lugol's iodine preparation of faeces show large number of pus cells, and erythrocytes.

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